

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I M Vishnu Vardhan Reddy(192572062) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Analytical Problem Solving

1. **Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.**

Introduction

A **hypervisor** is virtualization software that enables multiple virtual machines (VMs) to run on a single physical server by sharing hardware resources such as CPU, memory, storage, and network interfaces. Hypervisors improve hardware utilization, reduce infrastructure costs, and provide flexibility in cloud computing environments. Hypervisors are classified into **Type-1 (Bare-Metal)** and **Type-2 (Hosted)** hypervisors.

Type-1 Hypervisor (Bare-Metal)

A Type-1 hypervisor is installed directly on the physical hardware without requiring a host operating system. It manages hardware resources directly and provides virtual machines with secure access to CPU, memory, storage, and networking. Because there is no intermediate operating system, it offers high performance, better resource utilization, and stronger security.

Characteristics

- Installed directly on server hardware.
- Provides near-native performance.
- Supports enterprise cloud infrastructure.
- Suitable for large-scale virtualization.
- Offers high availability and fault tolerance.

Advantages

- Very low virtualization overhead.
- High scalability.

- Better security due to reduced attack surface.
- Supports live migration and clustering.
- Efficient CPU and memory scheduling.

Disadvantages

- Requires specialized installation and configuration.
- Higher deployment complexity.
- Enterprise licensing may increase cost.

Type-2 Hypervisor (Hosted)

A Type-2 hypervisor is installed on top of an existing operating system such as Windows or Linux. The host operating system controls the hardware, while the hypervisor manages the virtual machines. Since all hardware requests pass through the host OS, performance is lower than Type-1 hypervisors.

Characteristics

- Runs as an application.
- Suitable for desktop virtualization.
- Easy installation.
- Used mainly for testing and software development.

Advantages

- Simple installation.
- Easy to configure.
- Ideal for educational environments.
- Low hardware requirements.

Disadvantages

- Lower performance.
- Increased resource overhead.
- Security depends on host operating system.
- Not suitable for enterprise production environments.

Comparison Table

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Directly on hardware	On host operating system
Performance	Very High	Moderate
CPU Efficiency	Excellent	Lower
Memory Management	Efficient	Less efficient
Security	Very High	Moderate
Resource Overhead	Very Low	High
Scalability	Excellent	Limited
Enterprise Support	Yes	Limited
Live Migration	Supported	Limited
Typical Usage	Cloud Datacenters	Personal/Desktop Systems

Performance Analysis

Performance is one of the most important requirements for mission-critical applications such as banking systems, hospital management systems, airline reservation systems, and government portals.

Type-1 hypervisors directly interact with hardware, eliminating unnecessary software layers. This reduces CPU scheduling delays and improves memory management. Applications experience lower latency and higher throughput.

Type-2 hypervisors rely on the host operating system for hardware access. Every hardware request passes through

the host OS, increasing processing overhead and reducing performance.

Security Analysis

Security is essential for protecting sensitive enterprise applications.

Type-1 hypervisors have fewer software components, reducing the attack surface. They provide secure isolation between virtual machines and prevent unauthorized access to memory and storage.

Type-2 hypervisors inherit vulnerabilities from the host operating system. If the host OS is compromised, all hosted virtual machines may also be affected.

Manageability Analysis

Enterprise cloud environments require centralized monitoring and management.

Type-1 hypervisors support:

- Live Migration
- VM Snapshots
- High Availability (HA)
- Fault Tolerance (FT)
- Distributed Resource Scheduling (DRS)
- Centralized Monitoring
- Automatic Load Balancing

Type-2 hypervisors provide only basic management features suitable for testing environments.

Justification

Mission-critical applications require high availability, maximum security, and excellent performance. Therefore, **Type-1 hypervisors are the preferred choice** because they provide enterprise-grade reliability, strong isolation, and efficient resource utilization.

2. **A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.**

Introduction

Cloud providers often host applications from multiple organizations on the same physical hardware. For example, finance, healthcare, and student portals may share a common server. Since these applications process confidential information, strong isolation mechanisms are required.

Virtualization architecture enables multiple virtual machines to operate independently while sharing the same physical resources.

Virtualization Architecture

The hypervisor creates multiple independent virtual machines.

Each VM contains:

- Virtual CPU
- Virtual Memory
- Virtual Storage
- Virtual Network Interface
- Separate Guest Operating System
- Independent Applications

Even though all VMs share the same physical hardware, they remain logically isolated.

Compute Isolation

Each virtual machine receives dedicated virtual CPU resources.

Benefits:

- Prevents CPU interference.
- Fair resource allocation.
- Better workload performance.
- Reduced contention.

Memory Isolation

The hypervisor allocates separate memory regions for every virtual machine.

Benefits:

- Prevents memory leakage.
- Eliminates unauthorized memory access.
- Improves application security.

Storage Isolation

Each workload receives its own virtual storage.

Benefits:

- Independent virtual disks.
- Secure storage access.
- Data confidentiality.
- Easier backup and recovery.

Network Isolation

Software Defined Networking (SDN), VLANs, and virtual firewalls separate network traffic.

Benefits:

- Secure communication.
- Controlled access.
- Reduced network attacks.
- Better traffic management.

Resource Scheduling

The hypervisor continuously monitors:

- CPU utilization
- Memory utilization
- Storage utilization
- Network bandwidth

Resources are dynamically allocated according to workload demands.

3. **Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.**

Given Values

Resource	Utilization
CPU	92%
Memory	88%
Storage Latency	35 ms
VM Boot Time	170 s

CPU Analysis

A CPU utilization of **92%** indicates heavy processor usage.

Possible causes:

- Too many virtual machines

- CPU oversubscription
- Compute-intensive applications

Effects:

- Increased response time
- Scheduling delays
- Poor application performance

Memory Analysis

Memory utilization of **88%** indicates high RAM consumption.

Possible causes:

- Large number of active VMs
- Memory-intensive applications
- Memory overcommitment

Effects:

- Swapping
- Slow execution
- Reduced VM performance

Storage Analysis

Storage latency of **35 ms** is significantly higher than recommended values.

Possible causes:

- HDD storage
- Storage congestion
- Large number of read/write operations

Effects:

- Slow application startup
- Long VM boot time
- Database delays

VM Boot Time Analysis

An average boot time of **170 seconds** indicates poor infrastructure performance.

Likely reasons:

- Slow storage subsystem
- CPU bottleneck
- Memory pressure
- Network storage delays

Overall Bottlenecks

- CPU saturation
- Memory shortage
- High storage latency
- Slow disk subsystem
- Resource contention

Recommended Corrective Actions

CPU Improvements

- Add more processors.
- Reduce CPU oversubscription.
- Migrate VMs to less loaded hosts.

Memory Improvements

- Upgrade physical RAM.

- Optimize memory allocation.
- Remove unused VMs.

Storage Improvements

- Replace HDD with SSD/NVMe.
- Enable storage caching.
- Optimize storage I/O scheduling.

Virtualization Improvements

- Enable Dynamic Resource Scheduling.
- Use Live Migration.
- Implement Load Balancing.
- Monitor resource utilization continuously.

Expected Improvements

- Lower CPU utilization
- Faster VM boot time
- Reduced storage latency
- Improved application performance
- Better user experience

4. **Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.**

Introduction

A Software Defined Datacenter (SDDC) is a modern cloud infrastructure where compute, storage, and networking resources are virtualized and managed entirely through software. Unlike traditional datacenters, SDDCs automate resource provisioning, improve scalability, and reduce operational costs.

Traditional Datacenter

Characteristics:

- Physical servers
- Dedicated storage
- Hardware switches
- Manual configuration
- Slow deployment
- Limited scalability

Problems:

- High maintenance
- Low flexibility
- Long deployment time
- Poor resource utilization

Software Defined Datacenter

Characteristics:

- Virtualized servers
- Software Defined Storage (SDS)
- Software Defined Networking (SDN)
- Centralized management
- Automation
- Dynamic resource allocation

Compute Virtualization

Physical servers are divided into multiple virtual machines.

Advantages:

- Better utilization
- Fast provisioning
- Live migration
- Automatic scaling

Storage Virtualization

Multiple storage devices are combined into one logical storage pool.

Advantages:

- Flexible allocation
- Simplified management
- Faster backup
- Improved disaster recovery

Network Virtualization

Network resources are controlled through software.

Advantages:

- Rapid network creation
- Automated routing
- Better traffic management
- Improved security

Comparison

Traditional Datacenter Software Defined Datacenter

Manual configuration	Automated configuration
Fixed hardware	Virtual resources
Slow deployment	Rapid deployment
High maintenance	Lower maintenance
Limited scalability	Highly scalable
Low resource utilization	High utilization

Benefits

- Faster deployment
- Better agility
- Improved scalability
- Reduced operational cost
- Centralized monitoring
- High availability
- Disaster recovery
- Efficient resource utilization

Conclusion

Software Defined Datacenters significantly improve agility by virtualizing compute, storage, and networking resources. Automation and centralized management enable organizations to deploy and manage cloud services quickly and efficiently.

5. **A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.**

Introduction

Many organizations use multiple cloud providers to improve availability and reduce dependency on a single vendor. However, different providers often use different identity management systems, causing authentication failures and inconsistent access control.

Problem Analysis

Identity mismatches occur because:

- Different user databases
- Different authentication protocols
- Separate access policies
- Lack of trust relationships

Effects include:

- Login failures
- Duplicate accounts
- Poor user experience
- Security vulnerabilities
- Increased administration

Federated Identity Management (FIM)

Federated Identity Management enables users to authenticate with one trusted identity provider and access multiple cloud services without creating separate accounts.

Benefits:

- Centralized identity management
- Reduced password duplication
- Simplified administration
- Improved security

Single Sign-On (SSO)

SSO allows users to log in once and access multiple cloud applications.

Advantages:

- Better user experience
- Fewer passwords
- Reduced help-desk requests
- Faster authentication

Standard Federation Protocols

- **SAML 2.0** – Secure exchange of authentication information.
- **OAuth 2.0** – Secure authorization for applications.
- **OpenID Connect (OIDC)** – User authentication built on OAuth 2.0.

These standards allow different cloud providers to trust and verify user identities securely.

Privacy Protection

To protect sensitive user information:

- Encrypt identity data in transit and at rest.
- Share only the minimum required user attributes.
- Maintain audit logs for compliance.
- Apply Role-Based Access Control (RBAC).
- Use Multi-Factor Authentication (MFA).
- Regularly monitor identity-related activities.

Secure Identity Architecture

1. User logs in through a trusted Identity Provider (IdP).
2. IdP authenticates the user using MFA.
3. Identity information is securely shared using SAML/OIDC.
4. Cloud providers validate the identity token.
5. RBAC grants appropriate permissions.
6. All authentication events are logged for auditing.

Benefits

- Seamless access across multiple clouds.
- Strong authentication and authorization.
- Better privacy protection.
- Reduced identity duplication.
- Improved compliance with security regulations.
- Simplified administration and user management.

Conclusion

A secure multi-cloud federation should implement **Federated Identity Management (FIM), Single Sign-On (SSO), SAML 2.0, OAuth 2.0, OpenID Connect (OIDC), Role-Based Access Control (RBAC), and Multi-Factor Authentication (MFA)**. This approach ensures secure authentication, protects user privacy, and enables seamless access across multiple cloud platforms.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: